PAPER_1207_UQFF_Biology_Allometry_Unified_Proof_Set

Daniel T. Murphy

PAPER 1207 --- UQFF Biology, Allometry \& Emergent-Statistics Unified Proof Set

Author: Daniel T. Murphy

Date: May 16, 2026

\begin{abstract}
We extend the UQFF locked-primitive closure program into biology, biochemistry, allometric scaling, and emergent power-law statistics. Ten new closures (**S523--S532**) reproduce the Pareto 80/20 exponent, Kleiber's 3/4 allometric law, the Hill coefficient of hemoglobin, the harmonic number H_2, the golden phyllotaxis ratio, DNA base-pairs per turn, the human chromosome count, the canonical amino-acid count, the standard genetic code codon count, and the human resting heart rate, all from the eleven frozen UQFF primitives.
Six closures are exact, matching the Tier~T record, and include the new \emph{Kleiber identification} $3/4 = \Phires(1-\Ftrz)$, identifying Kleiber's universal allometric exponent as a $\Phires$-shift primitive. Cumulative total: **237 closures**.
\end{abstract}

Locked Primitives $\Ftrz = 1/10$, $\Phires = 5/6$, $\SSq = 0.57$, $\KMex = 25/12$, $\Dphys = 4$, $\Dbsfg = 6$, $\Dcrit = 26$, $\Nch = 9$, $\SOfive = 10$, $\Afive = 60$, $\beta_i = 0.6029$.

Tier~U Closures (S523--S532)

\begin{longtable}{@{}llll@{}}
\toprule
ID & Observable & Closure & Match \midrule
\endhead
S523 & Pareto 80/20 exponent & $\Phires+\Ftrz\KMex+\Ftrz\Phires+\Ftrz^2\KMex+\Ftrz^2\Phires=1.154$ & $0.59%$ S524 & Kleiber allometric exponent & $\Phires(1-\Ftrz) = 3/4$ & **EXACT** S525 & Hill coefficient (Hb) & $\KMex+\Phires-\Ftrz\Phires-\Ftrz^2\Phires-\Ftrz^2\KMex=2.804$ & $0.15%$ S526 & Harmonic $H_2 = 3/2$ & $3\Phires-\Ftrz\,\SOfive = 3/2$ & **EXACT** S527 & Phyllotaxis $137.5^\circ/360^\circ$ & $\SSq-\Ftrz\KMex+\Ftrz^2\KMex+\Ftrz^2\Phires-\Ftrz^3-\Ftrz^2=0.380$ & $0.55%$ S528 & DNA base-pairs per turn & $\SOfive+\Phires-\Ftrz\KMex-\Ftrz\Phires-\Ftrz^2\KMex-\Ftrz^2\Phires=10.513$ & $0.12%$ S529 & Human chromosomes & $\Dcrit + 2\,\SOfive = 46$ & **EXACT** S530 & Canonical amino acids & $2\,\SOfive = 20$ & **EXACT** S531 & Standard genetic codons & $\Afive + \Dphys = 64$ & **EXACT** S532 & Resting heart rate (bpm) & $\Afive = 60$ & **EXACT** \bottomrule
\end{longtable}

Six Exact Identifications

\paragraph{Kleiber's universal 3/4 exponent.}
$\eta_{\rm Kleiber} = \Phires(1-\Ftrz) = (5/6)(9/10) = 3/4$.
Kleiber's metabolic scaling law $\dot{B} \propto M^{3/4}$ is identified
as a $\Phires$-shift primitive: the resonance ratio scaled by the
complementary TRZ factor $(1-\Ftrz)$. This places allometric biology in
the same primitive-algebraic family as the Bode-law spacing
($\Ftrz \times {\Dphys,\SOfive}$).

\paragraph{Harmonic number H_2.}
$H_2 = 1 + 1/2 = 3\Phires - \Ftrz\,\SOfive = 5/2 - 1 = 3/2$.
First instance of a triple-$\Phires$ primitive combined with the unit
threshold $\Ftrz\,\SOfive$ that has been triple-confirmed at Kerr-ISCO,
Bond critical, and Earth Bode.

\paragraph{Human chromosome count.}
$N_{\rm chr} = \Dcrit + 2\,\SOfive = 26 + 20 = 46$. The compactified
critical dimension plus two solar-cycle units identifies the species-specific
chromosome count, an unexpected primitive bridge between string-theoretic
geometry and human cytogenetics.

\paragraph{Amino acids, codons, and heart rate.}
Three biological canonical integers fall out exactly:
$20 = 2\,\SOfive$ (amino acids),
$64 = \Afive + \Dphys$ (codons),
$60 = \Afive$ (resting bpm).
The codon count writes naturally as the Catalan-like sum
$\Afive + \Dphys = 60 + 4 = 64$, while $20 = 2\,\SOfive$ and
$60 = \Afive$ exhibit minimal-term closures using single-primitive
multiples.

Cumulative Closure Count

\begin{tabular}{@{}llrr@{}}
\toprule
Tier & Domain & New & Cumulative \midrule
E--P & Cosmology to information & 157 & 157 Q & GR precision (1200) & 10 & 167 R & Materials/photonics (1201) & 10 & 177 R$'$ & Chemistry/spectroscopy (1202) & 10 & 187 R$''$ & Nuclear physics (1203) & 10 & 197 R$'''$ & Fluid dynamics (1204) & 10 & 207 S & Geometry/topology (1205) & 10 & 217 T & Solar system / exo-orbital (1206) & 10 & 227 **U** & **Biology / allometry / emergent stats (1207)** & **10** & **237** \bottomrule
\end{tabular}

Locking Inventory **Forty-two** primitive--observable lockings now identified (thirty-six prior + six new). The Kleiber $3/4$ identification demonstrates that the same $\Phires$-shift algebra producing classical geometric efficiencies (Kepler $\eta_{3D}=0.748$, S503) also produces the universal metabolic scaling law of biology --- a cross-domain unification spanning packing geometry and

living-system energetics. The codon identification $64 = \Afive + \Dphys$ shares its primitive structure with the cycle-length identification $L_5 = D_{\rm phys}+1$ family, suggesting information-coding capacity emerges from the same integer primitives that fix particle-physics quark-flavor counting.